

Robustness of high-fidelity Rydberg gates with single-site addressability

Michael H. Green,^{1,2} Jia Z. Holger,^{1,2} Jia M. Ahn,² Johannes H. Busch,¹ and R. Brian Stokely²¹Department of Physics, University of Kansas, Lawrence, Kansas 66045-5050, USA²Department of Chemistry, University of California, Riverside, California 92521, USA

(Received 27 April 2018; published 26 September 2018)

Controlled-atom systems, a gate set formulated with respect to atom motion by making use of the Rydberg blockade, is becoming the leading technique to realize quantum computation with high-fidelity gates, however, it is complicated by experimental constraints on pulse amplitudes and timings, as well as other physical constraints. Here we use the Rydberg blockade to implement a controlled-atom system, and we use a quantum pulse sequence to implement a controlled-atom system. We report here a comprehensive study of the robustness of quantum computation to the Rydberg blockade gate set, and specifically examine the robustness of the gate fidelity with respect to such experimental parameters: detuning, pulse sequence, and atom positions. We consider the robustness of quantum computation to the Rydberg blockade gate set, and we use a quantum pulse sequence to implement a controlled-atom system. We report here a comprehensive study of the robustness of quantum computation to the Rydberg blockade gate set, and we use a quantum pulse sequence to implement a controlled-atom system. We report here a comprehensive study of the robustness of quantum computation to the Rydberg blockade gate set, and we use a quantum pulse sequence to implement a controlled-atom system.

DOI: 10.1103/PhysRevA.98.033801

PACS numbers: 03.67.Lx, 03.67.Mn, 32.80.Bx, 32.80.Dg

I. INTRODUCTION

Rydberg atoms, with a special internal degree of freedom, provide an attractive platform for realizing quantum information processing, offering a strong microwave frequency interface, dense and effective atom-atom interactions that may be switched on and off with respect to time [1]. Moreover, they have been used for quantum gates with both individually addressable and non-individually addressable single atom sites [2] as well as with atomic ensembles [3,4]. These features provide strong quantum coherence and make use of the Rydberg blockade to generate controlled-atom (CNOT) interactions, which are logical gates that are typically defined by the interaction of the ground electronic states, resulting in a robustness to coupling and compensating single atom site errors and optical resources that enable experimental realization of the key theoretical concepts of the Rydberg blockade [5,6], as well as subsequent use of this to generate a quantum blockade trapped atom in other configurations [7–11]. Moreover, [12] also demonstrated a low-fidelity quantum-controlled-atom gate, based on the Rydberg blockade, parallel to this, several groups have developed the capability to form arrays of trapped atoms in optical lattices that are characterized by single-site occupancy and addressability [13–17], thereby opening the path to large-scale quantum information processing with atomic ensembles.

Despite these conceptual and experimental advances, the realization of high-fidelity quantum logic gates remains a challenging task, due to the significant challenges involved in accurately controlling and manipulating optically trapped atoms. Two-fold gates relying on individual use of dipole transitions, between states in Rydberg atoms have the potential of being fast but are subject to a number of intrinsic and extrinsic sources of error that are linked back to the selected states and the speed of

operation. An important source of intrinsic error specific to Rydberg gates is the lifetime of the atoms in the Rydberg state, which is limited either by decay from a number of channels, or by decay to the ground state [18–21]. For the non-individually addressable implementation of the Rydberg gate presented in Ref. [2], atomic ensembles can provide a significant role in limiting the fidelity [17,18]. The intrinsic and extrinsic factors limiting gate fidelity have been studied theoretically for the Rydberg gate schemes involving both multi- and single-atom [18,19], and for the non-individually addressable [22], experimentally optimized protocol [17,18]. These studies indicate that gates with errors of the order of 10^{-4} might be achieved with suitable choice of atom and gate fields. However, no study of the robustness of two-qubit gates with respect to errors has been made, although such robustness with regard to fluctuations of such intrinsic and extrinsic parameters is a critical element of experimental studies. In this work we extend the theoretical study of the robustness of both multi- and non-individually addressable gates to the general two-fold dipole transition, the standard pulse timing, pulse amplitude, and two-photon detuning.

Further studies are for quantum information processing in the realization of fast gates. While proposals have been made to mitigate the effect of addressability errors in Rydberg gates using selective passage techniques [23], the resulting gate sequence typically would be relatively long gate times of approximately 10^{-4} to 10^{-3} s [24], which is disadvantageous for quantum computation schemes, since this generally requires large numbers of gates. Proposals for achieving gates on non-individual two atoms have been reported in Ref. [25]. For the non-individually addressable protocol of Ref. [2], optimal control theory has been used to characterize limits on the achievable gate fidelity [26], corresponding to a “quantum speed limit” for performing the gate [27].

The structure of the present work, and in brief, follows an introduction to the Rydberg blockade gate set, followed by a review of the various experimental and theoretical studies, as well as their comparison.

¹These authors contributed equally.